

2. Iron is extracted from iron oxide inside the blast furnace.



(a)

iron ore

steel

limestone

hot air

slag

iron

coke

waste gases

Choose substances from the box to complete the following sentences.

[2]

Iron oxide, coke and ..... are the materials fed in at the **top** of the furnace.

The furnace is heated by burning the coke in .....

(b) Balance the symbol equation that represents the main reaction taking place inside the furnace.

[1]



- (c) Some of the impurities are removed from the furnace as slag. Slag contains calcium silicate which has the chemical formula  $\text{CaSiO}_3$ .

Tick (✓) the box next to the calculation used to find the correct relative formula mass of calcium silicate,  $\text{CaSiO}_3$ . [1]

$$A_r(\text{Ca}) = 40$$

$$A_r(\text{Si}) = 28$$

$$A_r(\text{O}) = 16$$

$$40 + 28 + 16$$

☐

$$40 + 40 + 40 + 28 + 28 + 28 + 16 + 16 + 16$$

☐

$$40 + 28 + 16 + 16 + 16$$

☐

$$40 + 16 + 16 + 16$$

☐

$$40 + 28 + 28 + 28 + 16 + 16 + 16$$

☐

- (d) 820 tonnes of iron was extracted from 1 750 tonnes of iron ore. Calculate the percentage of iron in the ore. Give your answer to **one** decimal place. [2]

Percentage = ..... %

